Tran Ngoc Tuong

AI Engineer

Ha Noi, Vietnam

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Summary

AI Engineer with 5 years of experience developing and deploying AI systems. My expertise includes 4 years specializing in Computer Vision and Image Processing, 3 years working on Natural Language Processing/LLM projects.

I have a solid background in mathematics and informatics, including linear algebra, calculus, probability theory, and programming.

Education

2015 - 2020 Applied Mathematics and Informatics of Engineering at **HUST** GPA: 311/4.0

Skills

Operating System	Debian, Ubuntu, CentOS
Data Storage	PostgreSQL, MongoDB, MySQL, Milvus, Neo4J, etc.
Programming Language	Python, C/C++, OpenCV, Bash
Library and Framework	Pandas, Scikit-learn, Tensorflow, Pytorch, Transformers, Model compression (ONNX, TFLite, TensorRT), TF-serving, LLMDeploy, Huggingface, etc.
Version Control	Git, DVC
MLOps & Edge Devices	FastAPI, gRPC, CI/CD, Docker, Kubernetes, NVIDIA Jetson Nano
Languague	Vietnamese (native), English (TOIEC: 750)

Work Experience

AI Research Engineer at OmiNext JSC. Sept 2025 — Present

- Research and develop end-to-end OmiKG system, a biomedical ontology and knowledge graph architecture for mechanistic root cause analysis, achieving average score > GPT-5.2 and RAG system.
- Contributed to a research paper accepted for Oral Presentation at the AMIA 2026 Annual Symposium.
- Lead researcher - design and formulate the evaluation methodology for the research paper.
- Collaborate with research team members and domain experts to advance project objectives.

AI Computer Vision Engineer/ Data Scientist at Kalapa JSC. Sept 2024 — Sept 2025

- Enhance and compress models to detect fraudulent IDs and passports and develop a face anti-spoofing module to prevent presentation attacks on mobile cameras in the Kalapa eKYC system.
- Develop an end-to-end system that using LLMs, VLMs, and traditional OCR methods to extract information from unstructured documents, exceeding 50 pages and processing speeds of 5s/page.

AI Computer Vision Engineer at Eastgate Software Jun 2021 – Sept 2024

- Develop and optimize a video analytics system for HKeMobility including traffic flow forecast, speed estimation, queue length estimation, and accident detection on RTSP streams using Ultralytics YOLO.
- Build an AI-powered document processing platform for medical and veterinary report analysis, fine-tuning LayoutXLM, OCR, and NER models to achieve $F1 \geq 92\%$ accuracy while surpassing existing deep learning solutions by up to 39.51% through Language Independent Entity Linking.

- Built an end-to-end RAG-driven ESG, CSR, and Sustainability Reporting system with automated data extraction, report generation, predictive scoring, and model retraining using Azure OpenAI, Azure Blob Storage, Form Recognizer, and LLM caching.

AI Computer Vision Engineer at VNPT AI Sept 2019 - May 2021

- Key contributor in developing and optimizing an intelligent video analytics system for surveillance cameras (VNPT SmartVision) including AI engines for action detection, dangerous weapon detection (e.g., firearms), crowd detection, and camera tampering or occlusion detection, ...
- Develop an AI Engine for detecting special characteristics in the VNPT eKYC system.

Projects & self-study experience

Re-implementing Machine Learning algorithms from scratch Github profile

In addition to my professional work, I dedicate time to gaining a deeper understanding of core algorithms by re-implementing them from scratch, including OCR models (CRNN, Differentiable Binarization, etc.); Computer Vision algorithms such as object detection, segmentation, and human pose estimation (YOLO series, CenterNet, OpenPifPaf, etc); and Vision Large Language Models (CLIP, BLIP, ...)

Face Recognition - Active Learning, VIASM Project profile

The Vietnam Institute for Advanced Study in Mathematics (VIASM) collaborated with the Center for Supporting Development of Science and Technology (CENSTED) to organize training courses under the Active Learning project, an initiative founded by alumni of the Vietnam Education Foundation (VEF) scholarship program, with support from the U.S. Embassy in Hanoi. I successfully completed the program and was awarded a certificate, presenting a final project on Face Recognition.

Language independent Entity Linking Github profile

Developed a language-independent entity linking algorithm for Key-Value relation extraction in document understanding, achieving state-of-the-art F1-scores across 8 languages on the RFUND benchmark — significantly outperforming transformer-based models such as LayoutXLM, LayoutLMv3 and PEO.

Other projects

- Agentic Document Extraction (ADE): pulls specific data fields from parsed documents using schema-based extraction.
- Local Citation Recommendation (LCR): finetune BERT models for reranking stage to query relevant articles to the given citation context.
- Stamp Similarity: the project aims to predict the similarity between two stamps based on 2D image correlation and CNN.
- Image Dehazing: Haze removal and image enhancement for cameras (fog, rainy, smoke, backlighting, haze outdoors ...)
- Personal protective equipment (PPE): build a system to detect and monitor the personal protective equipment of workers at construction sites.

Certifications

- Certificated ML/DL course sponsored by Viet Nam Institute for Advanced Study in Mathematics
- MLOps — Machine Learning Operations Specicalization by Coursera
- Certificated Kubernetes for the Absolute Beginners-Hands-on by Udemy

Honors & awards

- Selected paper for oral presentation at the AMIA 2026 Annual Symposium (Dallas, TX, Nov 2026)
- Best applicability project prize of scientific research contest for students (22 May, 2019)
- Selected for the AI Fresher Program at VNPT-IT (2019)
- Top 15/30 Final Round - CODEWAR 2019 programming contest organized by FPT (24 Oct, 2019)
- Encouragement Award, District Mathematics Competition for Outstanding Students (2009)

Activities

- Collaborated with domain experts at National Institute of Nutrition (27 Jan, 2026)
- Teaching Assistant, AI Pioneers – AI Adoption Training Program at Ominext Jsc. (20 Nov, 2025)
- Attended the KIT–ASEAN 2025 Scientific Conference at Hanoi Medical University (12 Dec, 2025)
- Participated in the Student Research Conference at HUST (22 May, 2019).

Publications

Nguyen Nhung, Ngoc Khuc, Long Phi, Thuy Tran, Tuong Tran, Huong Tran, Huan Khuc, Ngan Hoang, and Dung Tran (2026). "OmiKG: An Ontology and Knowledge Graph for Mechanistic Root Cause Analysis in Functional Medicine". In: accepted for oral presentation at the AMIA 2026 Annual Symposium.